# Physics-Informed Neural Networks for 2D Elasticity: Application to a Cantilever Beam Benchmark

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November 2025

**Abstract**

This report investigates the use of Physics-Informed Neural Networks (PINNs) to approximate the plane-stress response of a cantilever beam subjected to transverse loading, and benchmarks the resulting surrogate against a finite element reference solution. A fully connected PINN is trained in strong form to represent the in-plane displacement field, using a multi-term loss that combines equilibrium residuals, clamped and traction-free boundary conditions, and beam-level constraints such as moment balance, tip deflection and neutral axis position.

In a first, physics-only stage, the network is trained without any simulation data. The resulting surrogate already reproduces the global beam response with errors of only a few percent with respect to both Euler–Bernoulli beam theory and the OptiStruct finite element solution. A second, data-assisted stage fine-tunes the model using a hybrid loss that adds a misfit to nodal displacements and axial stresses from FEA. This preserves the excellent agreement on beam-level quantities while noticeably improving the accuracy of the axial stress field near the clamped edge.

Error maps highlight typical limitations of strong-form PINNs in high-gradient regions, linked to the distribution of collocation points and the balance between competing loss terms. Overall, the study shows that moderately sized PINNs with carefully designed physics-based losses can act as accurate, differentiable surrogates complementary to classical FEA, and points towards extensions to parameterized and graph-based models for more complex geometries and loading scenarios.

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# 1 Introduction

Numerical simulation plays a central role in the analysis of mechanical structures, enabling the prediction of displacement, strain, and stress fields across a wide range of engineering applications. Among the available techniques, the Finite Element Method (FEM) has established itself as the standard tool due to its robustness and ability to handle complex geometries. Nevertheless, the growing demand for optimization, parametric studies, and rapid evaluations highlights the limitations of classical approaches when large numbers of simulations are required.

In response to these challenges, the integration of machine learning techniques into mechanical simulation has attracted increasing attention, particularly for the development of surrogate models capable of accelerating or complementing traditional numerical methods. This trend falls within the broader field of Scientific Machine Learning, which seeks to combine physical modeling with data-driven strategies to improve generalization and reduce dependence on extensive datasets. Such hybrid approaches also open promising directions toward faster and more flexible simulation tools, with potential relevance for design iteration, large-scale optimization, and real-time digital twin applications.

Introduced by Raissi et al. (2019), Physics-Informed Neural Networks (PINNs) embed the governing equations and boundary conditions directly into the loss function of a neural network, enabling the computation of physically consistent solutions without relying on a mesh. A broad review by Karniadakis et al. (2021) summarizes the conceptual foundations of physics-informed learning and its early developments across scientific computing. More recently, the rapid expansion of PINN methodologies has been synthesized in the comprehensive review by Luo et al. (2025), which maps the latest advances in architectures, sampling strategies, and loss formulations, and highlights how physics-informed learning is evolving toward increasingly sophisticated and versatile approaches. Together, these contributions illustrate the progression of PINNs from a foundational concept to a rapidly developing research direction, reinforcing the relevance of examining their behavior on controlled mechanical benchmarks.

Studying the application of PINNs to a simple mechanical problem therefore provides a valuable opportunity to understand the fundamental mechanisms underlying physics-informed learning. This work adopts such an exploratory perspective by examining how a PINN can approximate the solution of a linear elasticity problem and by analyzing the characteristics that emerge from this formulation, which links physical equations and automatic differentiation.

The objective of this project is to evaluate the ability of a PINN to reconstruct the mechanical fields of a cantilever beam under transverse loading. The study focuses on the quality of the approximation, the behavior of the physical loss during training, and the specific challenges associated with enforcing elasticity equations and boundary conditions. A finite element solution is used as a reference to situate the performance of the PINN within a controlled setting. Understanding the behavior of such models on canonical problems constitutes a meaningful step toward their future deployment within more complex engineering workflows.

The methodology consists in applying a PINN to a two-dimensional linear elasticity problem defined on a cantilever beam. The model enforces the equilibrium equations and boundary conditions directly within the loss function and is trained using collocation points distributed throughout the domain and along its boundary. The analysis focuses on the predicted displacement and stress fields, as well as on loss convergence. The subsequent sections detail the formulation, implementation, and training parameters employed in this study.

This report is organized as follows. The first section introduces the mechanical problem and associated assumptions. The next section describes the neural network architecture used to construct the physics-informed model. The formulation of the PINN for the cantilever beam is then presented, followed by the numerical results. A comparison with a finite element reference solution is provided thereafter, leading to a discussion of the observations. Finally, a conclusion summarizes the main findings of this work.

## 2 Problem Definition

We consider a two-dimensional linear elastic cantilever beam in bending, modeled in plane stress. The mathematical formulation follows classical references in elasticity and FEM (Fish and Belytschko [12], Hughes [10]) as well as beam theory (Ugural and Fenster [11]). The problem is static, with no body forces.

### 2.1 Geometry and Material Properties

The domain $\Omega \subset \mathbb{R}^2$ represents the mid-plane of a rectangular cantilever beam. We choose a Cartesian coordinate system such that the origin is located at the bottom-left clamped corner. The coordinates are defined as

$$x \in [0, L], \qquad y \in [0, h],$$

with

$$L = 500 \text{ mm}, \qquad h = 50 \text{ mm}.$$

In the imported mesh, the beam therefore extends from $x = 0$ to $x = 500$ mm and from $y = 0$ (bottom fibre) to $y = 50$ mm (top fibre). The out-of-plane thickness is denoted by $t$ and is taken as constant along the beam.

The material is homogeneous, isotropic and linear elastic, with:

- Young's modulus: $E = 55{,}000$ MPa,
- Poisson's ratio: $\nu = 0.26$,
- thickness: $t = 2.0$ mm.

The second moment of area of the rectangular cross-section about its centroidal axis ($y = h/2$) is:

$$I = \frac{t\,h^3}{12}. \tag{2.1}$$

A plane stress assumption is used, appropriate for thin plates and beams where $\sigma_{zz} \approx 0$ (see, e.g., [12, 10]).

### 2.2 Governing Equations of Linear Elasticity

Let $\mathbf{u} = (u, v)^\top$ be the in-plane displacement field. The static equilibrium equations without body forces are:

$$\nabla \cdot \boldsymbol{\sigma} = \mathbf{0} \quad \text{in } \Omega. \tag{2.2}$$

In Cartesian coordinates $(x, y)$, Eq. (2.2) expands to

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} = 0, \tag{2.3}$$

$$\frac{\partial \sigma_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} = 0. \tag{2.4}$$

Under the small-strain assumption, the strain components are

$$\varepsilon_{xx} = \frac{\partial u}{\partial x}, \tag{2.5}$$

$$\varepsilon_{yy} = \frac{\partial v}{\partial y}, \tag{2.6}$$

$$\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}. \tag{2.7}$$

For an isotropic material under plane stress, Hooke's law gives (see, e.g., [12, 10]):

$$\sigma_{xx} = \frac{E}{1-\nu^2} \left(\varepsilon_{xx} + \nu\, \varepsilon_{yy}\right), \tag{2.8}$$

$$\sigma_{yy} = \frac{E}{1-\nu^2} \left(\nu\, \varepsilon_{xx} + \varepsilon_{yy}\right), \tag{2.9}$$

$$\sigma_{xy} = \frac{E}{2(1+\nu)} \gamma_{xy}. \tag{2.10}$$

Equations (2.3)–(2.10) define the PDE system to be enforced by the PINN.

## 2.3 Boundary Conditions

The boundary is decomposed into

$$\partial\Omega = \Gamma_L \cup \Gamma_R \cup \Gamma_T \cup \Gamma_B,$$

with:

$$\begin{aligned}
\Gamma_L &= \{(x, y) \,|\, x = 0,\ y \in [0, h]\},\\
\Gamma_R &= \{(x, y) \,|\, x = L,\ y \in [0, h]\},\\
\Gamma_T &= \{(x, y) \,|\, y = h,\ x \in [0, L]\},\\
\Gamma_B &= \{(x, y) \,|\, y = 0,\ x \in [0, L]\}.
\end{aligned}$$

**Clamped boundary at $x = 0$ :**

The left side is fully fixed:

$$\mathbf{u} = \mathbf{0} \quad \text{on } \Gamma_L. \tag{2.11}$$

**Free top and bottom edges :**

On the top and bottom edges, the traction is zero:

$$\boldsymbol{\sigma}\,\mathbf{n} = \mathbf{0} \quad \text{on } \Gamma_T \cup \Gamma_B, \tag{2.12}$$

i.e.

$$\sigma_{yy} = 0, \qquad \sigma_{xy} = 0.$$

**Loaded right edge :**

A vertical resultant load $F_{\text{tot}}$ applied downward is modeled as:

$$\int_{\Gamma_R} \sigma_{xy}(L, y)\, t\, ds = -F_{\text{tot}}, \tag{2.13}$$

with

$$F_{\text{tot}} = 120 \text{ N}.$$

This condition is enforced through an integral constraint inside the PINN loss.

## 2.4 Analytical Reference

For comparison, the Euler–Bernoulli beam subjected to an end load $F_{\text{tot}}$ (Ugural and Fenster [11], see also beam PINN studies such as [13]) gives the bending moment:

$$M(x) = -F_{\text{tot}}(L - x), \tag{2.14}$$

and transverse deflection along the neutral axis:

$$v(x) = -\frac{F_{\text{tot}} x^2}{6EI}(3L - x). \tag{2.15}$$

The corresponding tip deflection is:

$$v(L) = -\frac{F_{\text{tot}} L^3}{3EI}. \tag{2.16}$$

In the present coordinate system, $y \in [0, h]$ is measured from the bottom edge, so that the neutral axis is located at $y = h/2$. The bending stress distribution is therefore

$$\sigma_{xx}(x, y) = -\frac{M(x)\,\big(y - h/2\big)}{I}. \tag{2.17}$$

The maximum normal stress in bending occurs at the fixed end ($x = 0$) and at the extreme fibres $y = 0$ or $y = h$, at a distance $h/2$ from the neutral axis, yielding

$$\sigma_{xx}^{\max} = \frac{F_{\mathrm{tot}} L\, h}{2I}, \tag{2.18}$$

up to a sign depending on whether the top or bottom fibre is considered. These expressions provide reference quantities used later to benchmark the PINN.

In summary, the problem consists in finding the displacement field $\mathbf{u} = (u, v)$ and the associated stress field $\boldsymbol{\sigma}$ that satisfy Eqs. (2.3)–(2.10) in $\Omega$ together with the boundary conditions (2.11)–(2.13).

# 3 PINN Architecture

Physics-Informed Neural Networks (PINNs) are neural architectures designed to solve PDEs by embedding the governing equations of a physical system directly into the training objective. Introduced by Raissi, Perdikaris and Karniadakis [4], PINNs have rapidly become a central component of Scientific Machine Learning (SciML). Their appeal lies in the fact that the neural network does not simply learn from data: it is constrained to satisfy the underlying partial differential equations (PDEs), thus combining the expressive power of deep learning with the structure of continuum mechanics. This section reviews the fundamental principles of PINNs, before applying them to the cantilever beam problem in Section 4.

## 3.1 Neural network representation of the displacement field

Fully-connected feed-forward neural networks (MLPs) are the standard backbone of PINNs. They map an input vector to an output vector through successive affine transformations and nonlinear activation functions. For a layer $l$, the output is given by

$$\mathbf{z}^{(l)} = \sigma^{(l)}\big(\mathbf{W}^{(l)}\mathbf{z}^{(l-1)} + \mathbf{b}^{(l)}\big), \tag{3.1}$$

where $\mathbf{W}^{(l)}$ and $\mathbf{b}^{(l)}$ are the weights and biases of layer $l$, and $\sigma^{(l)}$ is a nonlinear activation such as tanh or SiLU. Stacking such layers defines a high-dimensional nonlinear mapping

$$f_\theta : \mathbb{R}^2 \to \mathbb{R}^2, \qquad (x, y) \mapsto (u_\theta(x, y), v_\theta(x, y)), \tag{3.2}$$

with trainable parameters $\theta = \{\mathbf{W}^{(l)}, \mathbf{b}^{(l)}\}_l$.

Neural networks are universal function approximators: by the classical results of Cybenko [1] and Hornik [2], they can approximate any sufficiently smooth function on a compact domain. This property is particularly relevant in solid mechanics, where the displacement field is smooth and differentiable. Consequently, MLPs are well suited to represent the components of $(u, v)$ and their spatial derivatives.

To improve conditioning and numerical stability, all physical coordinates are first mapped to the normalized square $[-1,1]^2$ through an affine transformation,

$$(\xi_1, \xi_2) = T(x, y), \tag{3.3}$$

as implemented in the functions `norm_xy` and `phys_to_xn`. The neural network therefore operates in a dimensionless and scale-free space, which is standard practice in PINN implementations.

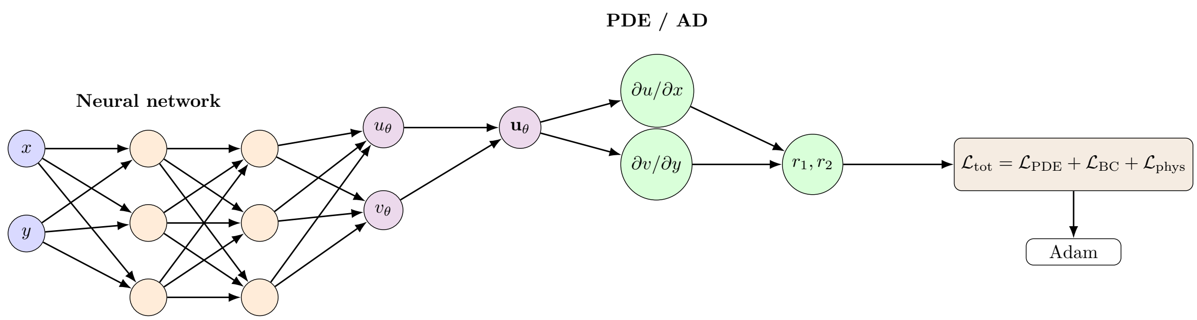


Figure 1: Illustration of the physics-informed neural network used in this work. A feed-forward neural network maps spatial coordinates $(x, y)$ (or their normalized counterparts $(\xi_1, \xi_2)$) to the displacement field $(u_\theta, v_\theta)$. Automatic differentiation provides the derivatives required to evaluate the PDE residuals $r_1, r_2$, which, together with boundary-condition and beam-theory priors, define the total loss $\mathcal{L}_{\text{tot}}$.

A crucial benefit of representing the displacement field with a neural network is that the model remains fully differentiable. Unlike mesh-based numerical methods, where gradients are computed locally on each element, PINNs use automatic differentiation to obtain exact derivatives of any order with respect to the physical coordinates. This property makes the network a natural candidate for enforcing differential operators directly, without relying on numerical approximation of strains or stresses. Moreover, the smoothness of standard activation functions (e.g. tanh) aligns well with the regularity requirements of elasticity problems, where the solution typically belongs to $H^1$ or smoother function spaces.

### 3.2 Automatic Differentiation for PDE Residuals

The central idea of a PINN is to convert a neural network into a differentiable surrogate of the mechanical solution. This is achieved by enforcing that the neural approximation satisfies the PDE in the strong form. Given a displacement field $(u_\theta, v_\theta)$, PyTorch's automatic differentiation (AD) computes the derivatives:

$$u_{\theta,x},\ u_{\theta,y},\ v_{\theta,x},\ v_{\theta,y}, \tag{3.4}$$

from which the strain components follow directly:

$$\varepsilon_{xx} = u_{\theta,x}, \qquad \varepsilon_{yy} = v_{\theta,y}, \qquad \gamma_{xy} = u_{\theta,y} + v_{\theta,x}. \tag{3.5}$$

Using the plane-stress constitutive law (Section 2), one obtains the stresses $\sigma_{xx}, \sigma_{yy}, \sigma_{xy}$, and finally the PDE residuals from the equilibrium equations,

$$r_1 = \sigma_{xx,x} + \sigma_{xy,y}, \qquad r_2 = \sigma_{xy,x} + \sigma_{yy,y}. \tag{3.6}$$

This automatic construction of the PDE residuals is one of the most powerful aspects of PINNs: the network becomes a fully differentiable model of elasticity, capable of satisfying arbitrary PDE operators without requiring numerical differentiation, mesh construction, or finite element shape functions.

**Sampling considerations:** The construction of the residuals $r_1$ and $r_2$ requires the evaluation of the PDE at a finite set of collocation points. The choice of these points plays a central role in the accuracy of the PINN: interior points enforce the equilibrium equations, while boundary points encode the Dirichlet and Neumann conditions. In practice, a mixture of uniformly sampled points and structured nodes (e.g. a mesh or grid) helps stabilize training by providing both geometric coverage and regularity along boundaries. The sampling strategy adopted for the cantilever beam problem is described in Section 4, together with the exact collocation sets used in practice.

### 3.3 Physics-Informed Loss Construction

PINNs enforce the governing mechanics through a loss function formulated in the strong form of the PDE. Denoting by $\mathcal{N}[u, v]$ the differential operator describing the physical model (here, the Navier–Lamé operator under plane stress), the equation loss is defined as a least-squares functional evaluated over collocation points:

$$\mathcal{L}_{\text{eq}} = \frac{1}{N_{\text{eq}}} \sum_{i=1}^{N_{\text{eq}}} \left\| \mathcal{N}[u_\theta, v_\theta](x_i) \right\|^2, \tag{3.7}$$

where $\mathcal{N}[u_\theta, v_\theta](x_i)$ collects the two equilibrium residuals corresponding to Eqs. (2.3)–(2.4). A similar loss decomposition is used in recent PINN studies for elasticity, such as Kag and Gopinath [7].

Boundary conditions are incorporated similarly through additional loss terms. Dirichlet conditions (e.g., fixed displacements) are enforced by penalizing deviations from the imposed values:

$$\mathcal{L}_{\text{D}} = \frac{1}{N_{\text{D}}} \sum_{i=1}^{N_{\text{D}}} \|\mathbf{u}_\theta(x_i) - \mathbf{u}_{\text{BC}}(x_i)\|^2, \tag{3.8}$$

where $\mathbf{u}_\theta = (u_\theta, v_\theta)$ and $\mathbf{u}_{\text{BC}}$ denotes the imposed displacement vector. Traction conditions (Neumann BCs) are enforced by constraining stress components or global resultants.

The total loss is then a weighted sum:

$$\mathcal{L}_{\text{tot}} = \lambda_{\text{eq}} \mathcal{L}_{\text{eq}} + \lambda_{\text{D}} \mathcal{L}_{\text{D}} + \lambda_{\text{N}} \mathcal{L}_{\text{N}} + \lambda_{\text{phys}} \mathcal{L}_{\text{phys}}, \tag{3.9}$$

where $\mathcal{L}_{\text{phys}}$ may include additional, problem-specific constraints (e.g., bending moment balance, neutral axis constraints, regularity priors). The detailed loss formulation used for the cantilever beam is presented in Section 4.

Although the formulation above is general, its performance is highly sensitive to the relative weights $\lambda_{\text{eq}}, \lambda_{\text{D}}, \lambda_{\text{N}}$ appearing in the total loss. These coefficients balance the enforcement of the PDE against boundary conditions and additional physical constraints. Improper weighting may lead to boundary-condition leakage, gradient imbalance, or optimization collapse. Various adaptive strategies exist in the literature, including gradient-norm balancing and residual-based weight annealing, but in this work the weights are chosen empirically based on preliminary experiments, following observations similar to those reported by Wang et al. [6]. Their final values and justification are presented in Section 4, alongside the other implementation details.

### 3.4 Training Strategy and Practical Considerations

The parameters $\theta$ are optimized using stochastic gradient descent (Adam optimizer) with an initial learning rate of $10^{-3}$ and a `ReduceLROnPlateau` scheduler. Batches contain a mixture of mesh nodes and randomly sampled interior points, following recommendations for stabilizing PINN training in stiff PDEs.

The best-performing network (with the lowest value of $\mathcal{L}_{\text{tot}}$) is stored and used for post-processing. This approach, analogous to early stopping, helps prevent overfitting to specific loss components.

**Gradient-conflict mitigation:** In practice, the total loss $\mathcal{L}_{\text{tot}}$ is a sum of several heterogeneous terms: local PDE residuals, displacement boundary conditions, traction-free conditions, beam-theory priors (moment balance, neutral axis), and global tip-deflection constraints. Each of these terms induces its own gradient in parameter space, and these gradients may point in conflicting directions. For instance, enforcing the Navier–Lamé equations in the interior can locally degrade the satisfaction of the clamped-edge boundary conditions, or conversely, a strong emphasis on the tip deflection may bias the network towards fitting a global constraint at the expense of the local stress field. This phenomenon is analogous to gradient interference in multi-task learning, where different objectives compete during training.

To alleviate this issue, the present work adopts the PCGrad strategy proposed by Yu et al. [3], which performs gradient surgery before each optimizer step. Let $(\mathcal{L}_1, \ldots, \mathcal{L}_M)$ denote the list of loss components (already weighted by their coefficients $\lambda_m$). For each $\mathcal{L}_m$, the gradient $g_m = \nabla_\theta \mathcal{L}_m$ is first computed with respect to all trainable parameters (in practice using `torch.autograd.grad` with `allow_unused=True`, replacing unused entries by zeros). PCGrad then iteratively projects away components of $g_i$ that are in conflict with $g_j$ for $j \neq i$. A conflict is detected when the dot product $\langle g_i, g_j \rangle < 0$, meaning that decreasing $\mathcal{L}_i$ would increase $\mathcal{L}_j$. In that case, $g_i$ is updated as

$$g_i \leftarrow g_i - \frac{\langle g_i, g_j \rangle}{\|g_j\|_2^2}\, g_j, \tag{3.10}$$

which removes from $g_i$ the component that opposes $g_j$. After this projection step has been applied over all loss pairs, the modified gradients $\{\tilde{g}_m\}_{m=1}^{M}$ are averaged and used to update the parameters via Adam; this procedure is implemented in the function `pcgrad_step` in the training script.

Empirically, PCGrad reduces destructive interference between the different physics-based terms and leads to a more balanced enforcement of the PDE, boundary conditions and beam-level constraints. In particular, it helps stabilize the competition between the local equilibrium residuals and the global moment-balance and tip-deflection losses, which would otherwise tend to dominate or be suppressed depending on the chosen loss weights.

In summary, a PINN combines a smooth neural representation of the displacement field with automatic differentiation and physics-based loss terms. This mesh-free formulation provides a flexible framework for solving elasticity problems without resorting to element-level discretization. However, its effectiveness depends strongly on architecture choices, sampling strategies, and loss weighting. These implementation details are problem-specific and will be described precisely in Section 4, where the approach is applied to the cantilever beam benchmark.

### 3.5 Comparison with the Finite Element Method

Before applying the method, it is instructive to compare PINNs with the classical numerical framework used in solid mechanics, namely the Finite Element Method (FEM). The goal here is not to perform a detailed performance comparison (this will be done in the Discussion section)but rather to highlight the conceptual differences between the two approaches.

Although both FEM and PINNs aim to approximate the solution of elasticity equations, they do so based on fundamentally different principles.

**Functional framework:** In displacement-based FEM, the solution belongs to a Hilbert space (typically $H^1$) and is the unique minimizer of an energy functional. Existence and uniqueness follow from the Lax–Milgram theorem. In contrast, PINNs approximate the solution in the space of neural networks, which does not form a classical function space and does not guarantee uniqueness.

**Weak form vs. strong form:** FEM relies on the weak form of the PDE, ensuring stability and satisfying the variational structure of the physics. PINNs, in their standard formulation, use the strong form and enforce it in a least-squares sense at collocation points.

**Discretization vs. continuous representation:** Finite elements discretize the domain into elements and use local polynomial basis functions. PINNs provide a global, mesh-free representation of the solution: the network directly approximates the continuous function $u_\theta(x, y)$ over the entire domain.

**Numerical stability:** FEM is well-conditioned by construction due to its variational formulation. PINNs may exhibit conditioning issues, gradient imbalance, and difficulties enforcing boundary conditions, especially in high-stiffness PDEs such as linear elasticity (see, e.g., Wang et al. [6]).

**Solution interpretation:** FEM computes a solution that satisfies the PDE in the energy norm. PINNs compute the solution that minimizes a least-squares error of PDE residuals, which does not correspond to the classical functional associated with elasticity. In that sense, FEM can be viewed as a variationally consistent, energy-based solver, whereas PINNs act as mesh-free least-squares residual minimizers.

These differences motivate the enriched loss formulation used in Section 4 to mitigate the known limitations of strong-form PINNs in elasticity problems.

A more detailed comparison between FEM and PINNs,including conditioning, convergence properties, and the practical limitations of strong-form formulations in elasticity,is deferred to the Discussion section. The present subsection serves only to establish the conceptual contrast between the two numerical paradigms.

# 4 PINN Formulation for the Cantilever Beam

In this section, we detail how the governing equations of 2D linear elasticity and beam theory are embedded into the loss function of a physics-informed neural network (PINN) to approximate the displacement and stress fields in the cantilever beam. The formulation follows the general PINN framework introduced by Raissi et al. [4] and recent extensions to solid mechanics and beam problems [7, 13].

The neural network takes as input a pair of spatial coordinates

$$\boldsymbol{x} = (x, y) \in \Omega = (0, L) \times (0, h)$$

and outputs the in-plane displacement field

$$\mathbf{u}_\theta(\boldsymbol{x}) = \begin{bmatrix} u_\theta(x, y) \\ v_\theta(x, y) \end{bmatrix},$$

where $\theta$ denotes the trainable parameters of the multilayer perceptron introduced in Section 3. All spatial derivatives required to evaluate the residuals are obtained by automatic differentiation.

### 4.1 Domain Sampling and Collocation Points

In the context of PINNs, the term *collocation points* refers to the set of spatial points in the domain and on its boundary at which the residuals of the partial differential equations (PDEs) and boundary conditions are enforced. These points play a role analogous to integration or sampling points in classical numerical methods, but they do not require a background mesh *per se* [4, 9]. Although PINNs do not require a mesh in principle, in this work we leverage the existing finite element mesh as a convenient source of structured collocation points.

The computational domain is defined by the mid-plane of the beam, $\Omega = (0, L) \times (0, h)$, with $L = 500$ mm and $h = 50$ mm, as introduced in Section 2. The finite element mesh used for the reference solution provides a set of nodal coordinates

$$\{\boldsymbol{x}_i\}_{i=1}^{N_{\text{mesh}}} = \{(x_i, y_i)\}_{i=1}^{N_{\text{mesh}}} \subset \Omega,$$

which are reused as collocation points in the PINN training.

To improve conditioning and to decouple the network from the physical scale of the problem, all spatial coordinates are mapped to a normalized domain $\tilde{\Omega} = [-1, 1]^2$ via an affine transformation. Let $x_{\min}$, $x_{\max}$, $y_{\min}$, and $y_{\max}$ denote the extrema of the physical mesh nodes. The normalization and inverse mapping are defined as

$$\tilde{x} = 2\,\frac{x - x_{\min}}{x_{\max} - x_{\min}} - 1, \qquad \tilde{y} = 2\,\frac{y - y_{\min}}{y_{\max} - y_{\min}} - 1, \tag{4.1}$$

$$x = x_{\min} + \frac{\tilde{x} + 1}{2}\,(x_{\max} - x_{\min}), \qquad y = y_{\min} + \frac{\tilde{y} + 1}{2}\,(y_{\max} - y_{\min}), \tag{4.2}$$

so that the network effectively operates on normalized inputs $\tilde{\boldsymbol{x}} = (\tilde{x}, \tilde{y}) \in [-1, 1]^2$.

The set of collocation points used during training is composed of:

- **Mesh-based points:** All finite element nodes are mapped to normalized coordinates $\{\tilde{\boldsymbol{x}}_i\}_{i=1}^{N_{\text{mesh}}}$ and used as a fixed dataset. A PyTorch DataLoader iterates over these points in mini-batches.

- **Random interior points:** In addition, a large bank of interior points is generated by uniform sampling in $\tilde{\Omega}$,

  $$\tilde{\boldsymbol{x}}_i^{\Omega} \sim \mathcal{U}([-1, 1]^2), \qquad i = 1, \ldots, N_{\Omega},$$

  with $N_{\Omega} \approx 2 \times 10^4$. For each batch, a subset of these points is combined with the mesh nodes to form an augmented set of interior collocation points on which the PDE residuals are evaluated.

- **Boundary points:** Boundary segments are extracted from the mesh:
  - the clamped left edge $\Gamma_D = \{x_{\min}\} \times [0, h]$,
  - the loaded right edge $\Gamma_{\text{right}} = \{x_{\max}\} \times [0, h]$,

– the top and bottom free surfaces $\Gamma_{\text{top}} = [0, L] \times \{y_{\text{max}}\}$ and $\Gamma_{\text{bottom}} = [0, L] \times \{y_{\text{min}}\}$.

The corresponding sets of normalized points are denoted $\mathcal{X}_D$, $\mathcal{X}_{\text{right}}$, $\mathcal{X}_{\text{top}}$ and $\mathcal{X}_{\text{bottom}}$.

- **Cross-sectional slices:** To impose additional mechanical constraints (neutral axis and bending moment balance), the domain is discretized into $N_x$ vertical slices at abscissas $\{x_c\}$ and $N_y$ points along the thickness in each slice. This defines a grid $\{(x_c, y_j)\}$ used to approximate cross-sectional integrals of $\sigma_{xx}$.

- **Mid-plane line:** Finally, points along the mid-plane $y = h/2$ are sampled to enforce that the axial stress is small along the neutral axis.

All these sets are precomputed in physical coordinates and then mapped to normalized coordinates before being passed through the network during training.

## 4.2 PDE Residual Terms

Given the displacement field $\mathbf{u}_\theta = (u_\theta, v_\theta)$, the small-strain tensor in plane stress is defined as

$$\varepsilon_{xx} = \frac{\partial u_\theta}{\partial x}, \qquad \varepsilon_{yy} = \frac{\partial v_\theta}{\partial y}, \qquad \gamma_{xy} = \frac{\partial u_\theta}{\partial y} + \frac{\partial v_\theta}{\partial x}, \tag{4.3}$$

where all derivatives are obtained by automatic differentiation with respect to the physical coordinates $(x, y)$.

For an isotropic material in plane stress, the Cauchy stress components follow from Hooke's law [10]:

$$\sigma_{xx} = \frac{E}{1-\nu^2}\big(\varepsilon_{xx} + \nu\varepsilon_{yy}\big), \qquad \sigma_{yy} = \frac{E}{1-\nu^2}\big(\nu\varepsilon_{xx} + \varepsilon_{yy}\big), \qquad \sigma_{xy} = G\,\gamma_{xy}, \tag{4.4}$$

with $E$ Young's modulus, $\nu$ Poisson's ratio and $G = E/(2(1+\nu))$ the shear modulus. In the implementation, these relations are encoded in the function `sigma_plane_stress`.

In the absence of body forces, the equilibrium equations read

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} = 0, \qquad \frac{\partial \sigma_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} = 0 \quad \text{in } \Omega. \tag{4.5}$$

These are enforced in a least-squares sense at the interior collocation points. For each point $\boldsymbol{x}_i^\Omega$, the PINN residuals are

$$r_1(\boldsymbol{x}_i^\Omega; \theta) = \frac{\partial \sigma_{xx,\theta}}{\partial x} + \frac{\partial \sigma_{xy,\theta}}{\partial y}, \tag{4.6}$$

$$r_2(\boldsymbol{x}_i^\Omega; \theta) = \frac{\partial \sigma_{xy,\theta}}{\partial x} + \frac{\partial \sigma_{yy,\theta}}{\partial y}, \tag{4.7}$$

and the corresponding loss term is the mean-squared equilibrium residual

$$L_{\mathrm{PDE}}(\theta) = \frac{1}{N_\Omega} \sum_{i=1}^{N_\Omega} \Big( r_1(\boldsymbol{x}_i^\Omega;\theta)^2 + r_2(\boldsymbol{x}_i^\Omega;\theta)^2 \Big). \tag{4.8}$$

Because the problem is explicitly formulated in plane stress, we also penalize deviations from $\sigma_{yy} \approx 0$ in the interior:

$$L_{\sigma_{yy}}(\theta) = \frac{1}{N_\Omega} \sum_{i=1}^{N_\Omega} \sigma_{yy,\theta}(\boldsymbol{x}_i^\Omega)^2. \tag{4.9}$$

This interior penalty complements the traction-free conditions imposed on the top and bottom surfaces, and reinforces the plane-stress assumption throughout the domain.

Beyond these local PDE residuals, the implementation includes additional mechanical constraints derived from beam theory [11]:

**Neutral axis condition (cross-sectional equilibrium):** For each vertical slice at position $x_c$, the axial force over the cross-section should vanish:

$$\int_0^h \sigma_{xx}(x_c, y)\, \mathrm{d}y \approx 0. \tag{4.10}$$

This integral is approximated by numerical quadrature over the sampled points $\{(x_c, y_j)\}$, leading to a loss $L_{\mathrm{neutral}}(\theta)$ built from the squared cross-sectional resultants.

**Mid-plane neutral axis:** Along the mid-plane $y = h/2$, the axial stress is expected to be small, which is enforced by

$$L_{\mathrm{mid}}(\theta) = \frac{1}{N_{\mathrm{mid}}} \sum_{i=1}^{N_{\mathrm{mid}}} \sigma_{xx,\theta}(x_i, h/2)^2. \tag{4.11}$$

**Bending moment balance:** The bending moment resulting from $\sigma_{xx}$ in each slice at $x = x_c$ must match the theoretical bending moment produced by the tip load $F_{\mathrm{tot}}$. With the sign convention adopted in Section 2, a downward tip load is represented by a positive magnitude $F_{\mathrm{tot}} > 0$, and the internal bending moment reads

$$M(x_c) = \int_0^h \sigma_{xx}(x_c, y)\, (y - h/2)\, t\, \mathrm{d}y \approx -F_{\mathrm{tot}}\, (L - x_c), \tag{4.12}$$

where $t$ is the beam thickness. This leads to a loss term $L_{\mathrm{moment}}(\theta)$ built from the squared difference between the predicted and target bending moments.

**Beam prior on the transverse displacement:** For a cantilever beam of bending stiffness $EI$ subjected to a tip load $F_{\mathrm{tot}}$, the Euler–Bernoulli theory predicts

$$EI \, \frac{\mathrm{d}^2 v}{\mathrm{d}x^2}(x) = M(x) = -\, F_{\mathrm{tot}}(L - x). \tag{4.13}$$

In the present formulation, $v_\theta(x, y)$ represents the vertical component of the 2D displacement field. To connect it with the one-dimensional beam model, this relation is encoded as an additional prior on $v_\theta$ by penalizing the deviation between the second derivative $\partial^2 v_\theta / \partial x^2$, obtained via automatic differentiation, and the analytical quantity $M(x)/(EI)$ at randomly sampled points in the domain. This defines a loss term $L_{\mathrm{beam}}(\theta)$ that encourages the longitudinal curvature of $v_\theta$ to be consistent with the Euler–Bernoulli bending curve.

All these contributions act as physics-based regularizers that guide the PINN towards a displacement and stress field consistent with both 2D elasticity and classical beam theory [12, 11, 7].

## 4.3 Boundary Condition Losses

The boundary conditions of the cantilever beam are enforced via several loss terms evaluated at the corresponding boundary collocation points.

**Clamped left boundary:** On the left edge $\Gamma_D = \{0\} \times [0, h]$, the beam is fully clamped:

$$\mathbf{u} = \mathbf{0} \quad \text{on } \Gamma_D. \tag{4.14}$$

Let $\mathcal{X}_D = \{\boldsymbol{x}_j^D\}_{j=1}^{N_D}$ denote the set of points extracted from the left edge of the mesh. The clamped boundary loss is defined as

$$L_D(\theta) = \frac{1}{N_D} \sum_{j=1}^{N_D} \big( u_\theta(\boldsymbol{x}_j^D)^2 + v_\theta(\boldsymbol{x}_j^D)^2 \big). \tag{4.15}$$

In addition to the soft penalty (4.15), the clamped boundary condition is also enforced "hard" at the architecture level: the network multiplies its raw output by a factor that vanishes at the normalized coordinate $\tilde{x} = -1$. This guarantees exact satisfaction of the essential Dirichlet condition and significantly improves the stability of the training process.

**Traction-free top and bottom surfaces:** The top and bottom edges of the beam are traction-free. In terms of normal and shear stresses, this implies

$$\sigma_{yy} = 0, \qquad \sigma_{xy} = 0 \quad \text{on } \Gamma_{\mathrm{top}} \cup \Gamma_{\mathrm{bottom}}. \tag{4.16}$$

Let $\mathcal{X}_{\text{top}}$ and $\mathcal{X}_{\text{bottom}}$ denote the sets of collocation points along these boundaries. The corresponding loss is

$$\begin{aligned}L_{\text{free}}(\theta) =& \frac{1}{N_{\text{top}}} \sum_{\boldsymbol{x} \in \mathcal{X}_{\text{top}}} \left(\sigma_{yy,\theta}(\boldsymbol{x})^2 + \sigma_{xy,\theta}(\boldsymbol{x})^2\right) \\ &+ \frac{1}{N_{\text{bottom}}} \sum_{\boldsymbol{x} \in \mathcal{X}_{\text{bottom}}} \left(\sigma_{yy,\theta}(\boldsymbol{x})^2 + \sigma_{xy,\theta}(\boldsymbol{x})^2\right).\end{aligned} \tag{4.17}$$

**Resultant of the applied tip load:** The external loading is modeled as a resultant vertical force $F_{\text{tot}}$ applied at the right end of the beam. In the 2D formulation, this is encoded by enforcing that the shear traction integrated over the right boundary matches $F_{\text{tot}}$:

$$\int_{\Gamma_{\text{right}}} \sigma_{xy}(x_{\max}, y)\, t\, \mathrm{d}y \approx -F_{\text{tot}}, \tag{4.18}$$

with $t$ the thickness. In practice, the right edge is discretized by mesh nodes, from which midpoints and segment lengths $\Delta s_m$ are constructed. The predicted resultant is then approximated as

$$F_{\text{pred}} = \sum_m \sigma_{xy,\theta}(\boldsymbol{x}_m^{\text{right}})\, t\, \Delta s_m, \tag{4.19}$$

and the corresponding loss term is

$$L_{\text{res}}(\theta) = \left(F_{\text{pred}} - F_{\text{tot}}\right)^2. \tag{4.20}$$

**Tip deflection and uniformity:** The Euler–Bernoulli beam theory provides an analytical expression for the vertical tip deflection under a tip load $F_{\text{tot}}$,

$$v_{\text{tip}}^{(\text{EB})} = -\frac{F_{\text{tot}} L^3}{3EI}, \tag{4.21}$$

with $I = th^3/12$ the second moment of area of the cross-section ($t = 2$ mm, $h = 50$ mm, hence $I \approx 2.08 \times 10^4$ mm$^4$). With $F_{\text{tot}} > 0$ representing the magnitude of a downward tip load, $v_{\text{tip}}^{(\text{EB})}$ is negative. Rather than imposing the exact traction distribution at the tip, we constrain the mean vertical displacement on the right edge to match $v_{\text{tip}}^{(\text{EB})}$:

$$L_{\text{tip}}(\theta) = \left(\overline{v_\theta}\big|_{\Gamma_{\text{right}}} - v_{\text{tip}}^{(\text{EB})}\right)^2, \tag{4.22}$$

where $\overline{v_\theta}\big|_{\Gamma_{\text{right}}}$ denotes the average of $v_\theta$ over the right boundary collocation points.

Additionally, we encourage the vertical displacement to be approximately uniform along the right edge, by penalizing the variance of the deviation from the mean:

$$L_{\text{uniform}}(\theta) = \operatorname{Var}\Big(v_\theta(\boldsymbol{x}) - \overline{v_\theta}\big|_{\Gamma_{\text{right}}}\Big), \quad \boldsymbol{x} \in \Gamma_{\text{right}}. \tag{4.23}$$

### 4.4 Hyperparameters and Implementation Details

Beyond the local equilibrium residuals, slender beam problems in 2D elasticity are known to be highly sensitive to boundary conditions, stress resultants, and cross-sectional consistency constraints. For this reason, the loss function is constructed as a weighted sum of complementary terms: some enforce the governing PDEs, others impose boundary conditions, and several act as mechanical priors (beam-theory consistency, moment balance, neutral axis). These additional constraints greatly stabilize training and prevent unphysical displacement or stress modes while avoiding the need for a large number of collocation points.

All components described above are combined into a single scalar loss function of the form

$$\begin{aligned} L(\theta) = {} & W_{\text{PDE}} L_{\text{PDE}} + W_{\sigma_{yy}} L_{\sigma_{yy}} + W_D L_D + W_{\text{free}} L_{\text{free}} + W_{\text{res}} L_{\text{res}} \\ & + W_{\text{neutral}} L_{\text{neutral}} + W_{\text{mid}} L_{\text{mid}} + W_{\text{moment}} L_{\text{moment}} \\ & + W_{\text{beam}} L_{\text{beam}} + W_{\text{uniform}} L_{\text{uniform}} + W_{\text{tip}} L_{\text{tip}}, \end{aligned} \tag{4.24}$$

where the $W_\bullet$ are positive weights used to balance the contributions of the different terms. In the experiments reported in Section 5, we use

$$\begin{aligned} W_{\text{PDE}} &= 0.03, & W_{\sigma_{yy}} &= 0.0, & W_D &= 1.0, & W_{\text{free}} &= 0.1, \\ W_{\text{res}} &= 0.3, & W_{\text{neutral}} &= 0.01, & W_{\text{mid}} &= 0.0, & W_{\text{moment}} &= 0.02, \\ W_{\text{beam}} &= 0.02, & W_{\text{uniform}} &= 1.0, & W_{\text{tip}} &= 200.0. \end{aligned} \tag{4.25}$$

The terms $L_{\sigma_{yy}}$ and $L_{\text{mid}}$ were kept in the formulation for completeness but assigned zero weight in the final model, as preliminary tests showed that activating them did not improve convergence for this specific geometry.

These weights are treated as hyperparameters and were tuned empirically to obtain a good compromise between: (i) satisfaction of the PDE and plane-stress conditions, (ii) accurate enforcement of the clamped boundary, free surfaces and tip resultant, and (iii) consistency with the beam-theory predictions for bending moment and tip deflection. In particular, a relatively large value of $W_{\text{tip}}$ is used to encourage the PINN to reproduce the analytical Euler–Bernoulli tip deflection, while the other weights remain moderate so as not to overconstrain the displacement field. A more detailed discussion of the influence of these weights on the final solution is provided in Section 8, in connection with the analysis of the error fields and stress distributions.

The network architecture is a fully-connected multilayer perceptron with input dimension 2, output dimension 2 (the components of the displacement field), and three hidden layers:

$$[2,\ 64,\ 128,\ 64,\ 2], \tag{4.26}$$

with SiLU activation functions in the hidden layers and Xavier initialization for all weights. The parameters are stored in double precision, and all derivatives are computed via PyTorch's automatic differentiation.

Training is performed using the Adam optimizer with an initial learning rate $\eta_0 = 10^{-3}$ and a small weight decay, together with a `ReduceLROnPlateau` scheduler that decreases the learning rate by a factor 0.8 when the total loss (4.24) stagnates. The network is trained for 3000 epochs, with mini-batches built from the mesh nodes and augmented at each iteration by randomly sampled interior points. The best-performing model (in terms of the total loss (4.24)) is stored and used for the post-processing of displacement and stress fields presented in Section 5.

# 5 Results

This section presents the displacement and stress fields predicted by the PINN and contrasts them qualitatively with the finite element (FE) reference solution described in Section 6.1. The evolution of the different loss terms introduced in Sec. 4 is also discussed. Quantitative error measures are deferred to Section 6.

## 5.1 Predicted Displacement Fields

The trained PINN provides an approximation of the in-plane displacement field $\mathbf{u}_\theta = (u_x, u_y)$ over the rectangular domain $\Omega = (0, L) \times (0, h)$.

Figures 2 and 4 show the axial and transverse displacement components $u_x(x, y)$ and $u_y(x, y)$ predicted by the network. For comparison, the corresponding fields from the FE reference solution are plotted in Figures 3 and 5.

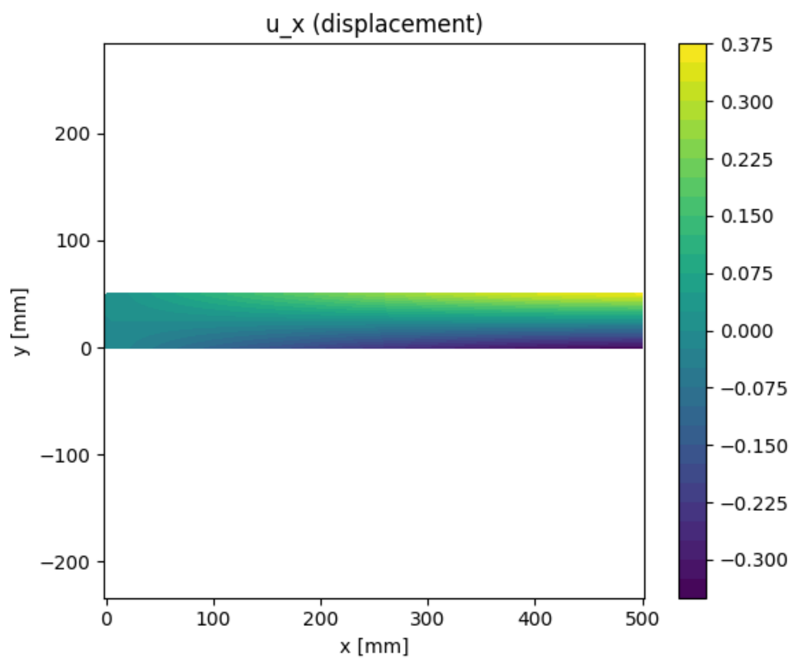


Figure 2: Axial displacement field $u_x(x, y)$ predicted by the PINN over the cantilever beam. The colour scale is in millimetres.

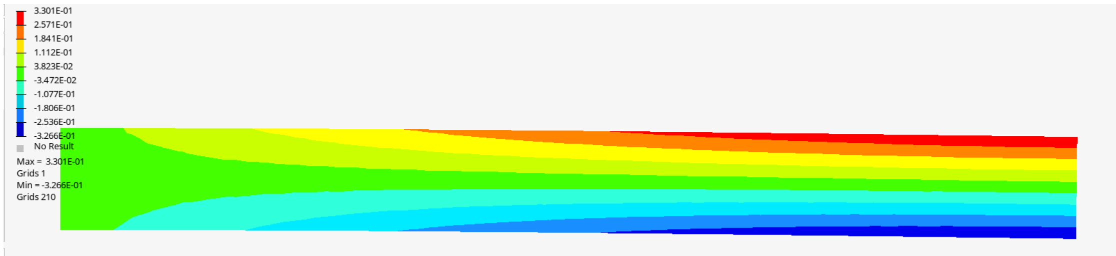


Figure 3: Axial displacement field $u_x(x, y)$ from the FE reference solution. The colour scale is in millimetres.

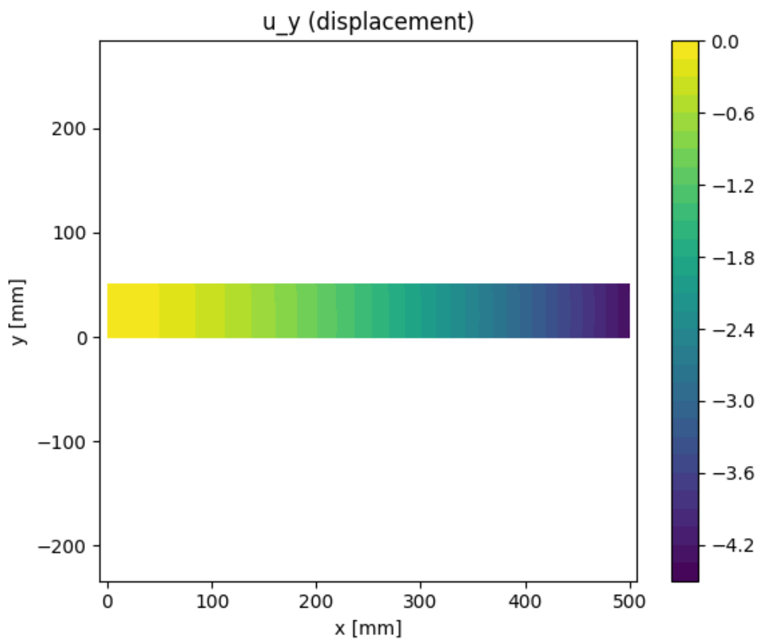


Figure 4: Transverse displacement field $u_y(x, y)$ (vertical deflection) predicted by the PINN. The left edge is clamped and the free end at $x = L$ exhibits the maximum downward deflection. The colour scale is in millimetres.

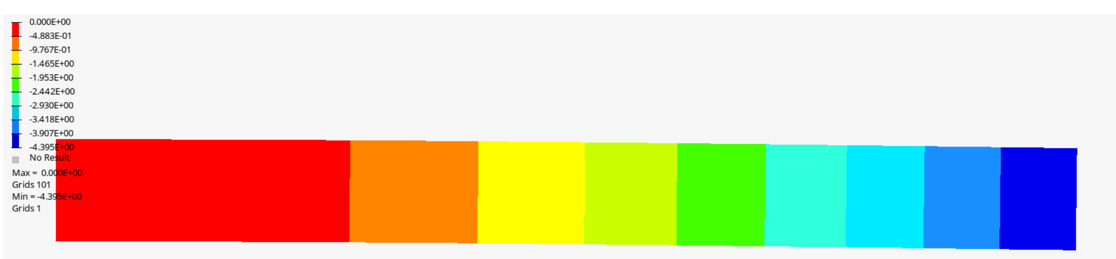


Figure 5: Transverse displacement field $u_y(x, y)$ from the FE reference solution. The colour scale is in millimetres.

The axial displacement $u_x$ in Fig. 2 exhibits the expected bending pattern. The field is approximately antisymmetric with respect to the mid-plane $y = h/2$, with extension on one side of the beam and compression on the other. The FE field in Fig. 3 shows a very similar distribution, with slightly sharper gradients near the clamped edge due to the piecewise-linear interpolation of the CQUAD4 elements.

The vertical displacement field $u_y$ (denoted $v$ in beam theory), shown in Fig. 4, is almost one-dimensional along the beam axis: level sets form smooth arcs from the clamped to the free end, with very little variation across the thickness. The FE contours in Fig. 5 display the same overall bending shape, with a slightly more pronounced curvature near the fixed support due to shear effects captured by the finite element formulation.

At the free end ($x = L$), the PINN predicts a mean right-edge deflection

$$v_{\text{right}}^{\text{PINN}} \approx -4.3616 \text{ mm},$$

while the analytical Euler–Bernoulli prediction for the applied loading is

$$v_{\text{theory}} \approx -4.3643 \text{ mm}.$$

The FE reference solution yields a tip deflection

$$v_{\text{right}}^{\text{FEA}} \approx -4.395 \text{ mm}.$$

The corresponding relative errors with respect to the Euler–Bernoulli value are therefore

$$\varepsilon_v^{\text{PINN}} = \frac{|v_{\text{right}}^{\text{PINN}} - v_{\text{theory}}|}{|v_{\text{theory}}|} \approx 0.06\%, \qquad \varepsilon_v^{\text{FEA}} = \frac{|v_{\text{right}}^{\text{FEA}} - v_{\text{theory}}|}{|v_{\text{theory}}|} \approx 0.70\%.$$

Both the PINN and the FE model thus reproduce the beam-theory compliance with excellent accuracy, the PINN being slightly closer to the analytical deflection at the tip. The remaining discrepancies between the methods are mainly associated with local stress and strain distributions, which are examined next.

## 5.2 Predicted Stress Fields

The normal stress component $\sigma_{xx}$ is recovered from the PINN displacement field using the linear elastic constitutive law. The resulting stress distribution is shown in Fig. 6, together with the FE counterpart in Fig. 7.

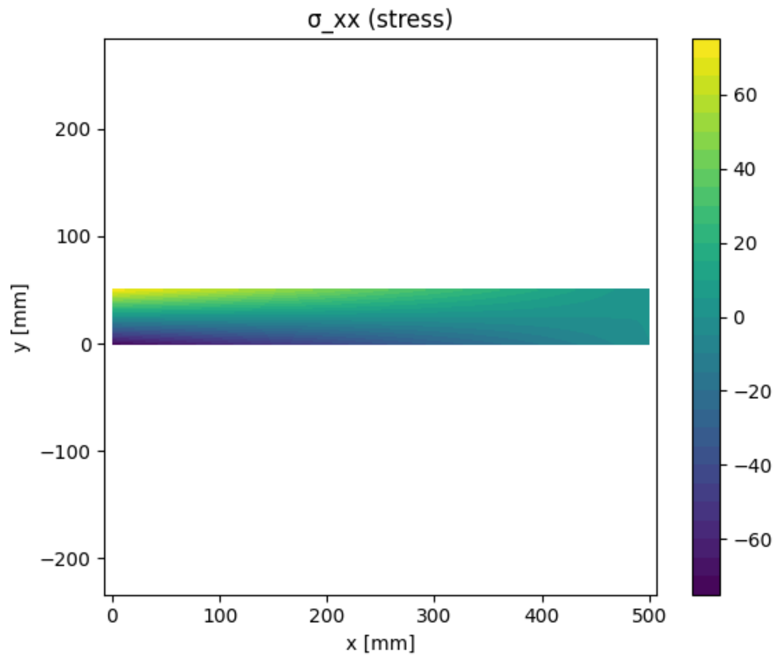


Figure 6: Normal stress field $\sigma_{xx}(x, y)$ predicted by the PINN. Units are MPa.

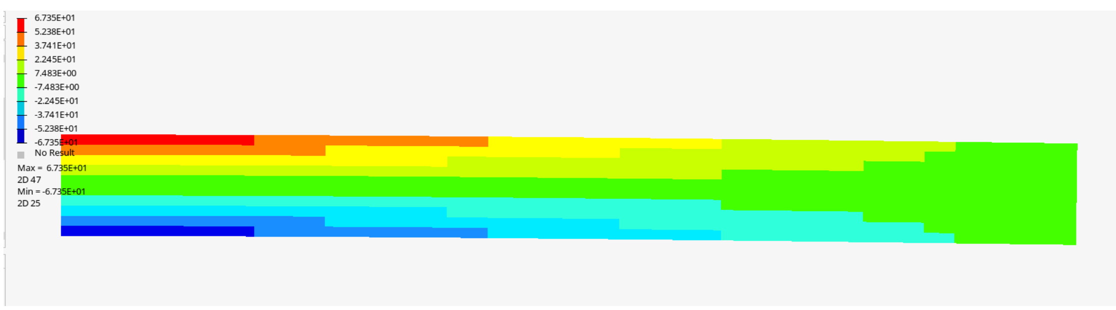


Figure 7: Normal stress field $\sigma_{xx}(x, y)$ obtained from the FE reference solution. Units are MPa.

As expected for a cantilever beam in bending, both the PINN and FE solutions display a nearly linear variation of $\sigma_{xx}$ through the thickness and an approximately uniform level along the span, away from the free end. The top and bottom fibres at the clamped edge carry the largest compressive and tensile stresses, while the mid-plane remains close to the neutral axis with small axial stress. The FE contours are almost perfectly straight in the vertical direction, whereas the PINN solution exhibits slightly curved bands near the fixed support, reflecting the smoothness of the neural network approximation and the residual PDE error in that region.

To quantify the agreement with beam theory, the maximum bending stress is evaluated along the clamped edge on the outermost fibre. Using the expression derived in Section 2, the theoretical maximum is

$$\sigma_{xx,\max}^{\text{theory}} \approx -72.012 \text{ MPa},$$

the negative sign corresponding to compression at the top fibre for a downward tip load. In terms of magnitude, this gives $|\sigma_{xx,\max}^{\text{theory}}| \approx 72.012$ MPa. The PINN prediction, sampled along the clamped-edge outer fibre, yields an extreme value of

$$|\sigma_{xx,\max}^{\text{PINN}}| \approx 71.609 \text{ MPa},$$

while the FE reference solution gives

$$|\sigma_{xx,\max}^{\text{FEA}}| \approx 67.35 \text{ MPa}.$$

Comparing magnitudes with respect to the Euler–Bernoulli value, the corresponding relative errors are

$$\varepsilon_{\sigma}^{\text{PINN}} = \frac{\left||\sigma_{xx,\max}^{\text{PINN}}| - |\sigma_{xx,\max}^{\text{theory}}|\right|}{|\sigma_{xx,\max}^{\text{theory}}|} \approx 0.6\%, \qquad \varepsilon_{\sigma}^{\text{FEA}} = \frac{\left||\sigma_{xx,\max}^{\text{FEA}}| - |\sigma_{xx,\max}^{\text{theory}}|\right|}{|\sigma_{xx,\max}^{\text{theory}}|} \approx 6.5\%.$$

Because of the relatively coarse mesh used in the present FE model, the FEA solution slightly underestimates the peak bending stress, whereas the PINN prediction remains much closer to the theoretical value.

In the PINN solution, the tensile and compressive extrema on the outer fibres are slightly asymmetric, a behaviour commonly observed in strong-form PINNs for elasticity, as the network does not enforce perfect antisymmetry of the stress field with respect to the neutral axis. Despite this, the predicted magnitude and thickness-wise profile of $\sigma_{xx}$ remain in very good agreement with both Euler–Bernoulli theory and the FE reference contours.

## 5.3 Loss Convergence

The evolution of the main loss contributions during training is shown in Fig. 8 on a logarithmic scale.

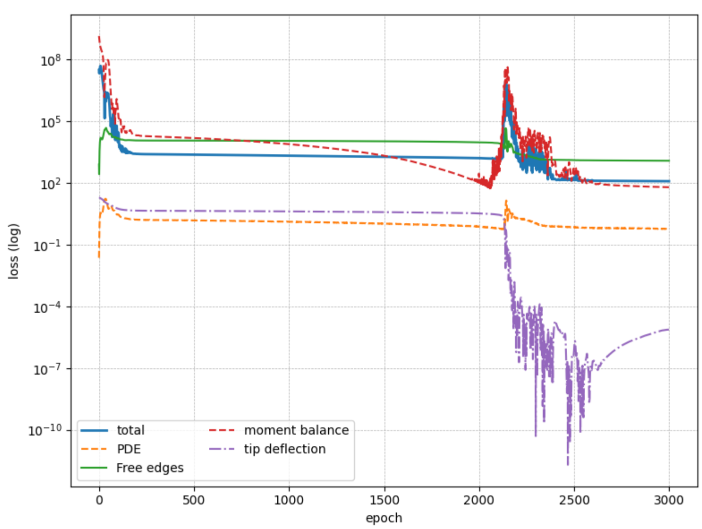


Figure 8: Convergence history of the total loss and selected components during training (logarithmic scale): PDE residual, free-edge boundary term, moment balance, and tip-deflection constraint.

The total loss exhibits an overall downward trend, with two pronounced transient phases where several components increase by a few orders of magnitude before relaxing again. The PDE residual loss $L_{\mathrm{PDE}}$ decreases rapidly in the early stages of training and then stabilizes on a plateau, indicating that the PINN learns a displacement field that satisfies the plane-stress equilibrium equations with good accuracy in the interior of the domain. The loss associated with the free-edge boundary conditions follows a similar pattern: after an initial transient, it reaches a moderate and nearly constant level, showing that the traction-free conditions are enforced reasonably well along the top and bottom surfaces.

The beam-level constraints, by contrast, converge more slowly and remain larger than the local terms. The moment-balance loss stays significantly above the PDE residual throughout training, showing that matching integral bending moments is more demanding than enforcing pointwise equilibrium. The tip-deflection loss $L_{\mathrm{tip}}$ remains roughly constant during the first training phase, then undergoes a sharp decrease around the second re-weighting stage, dropping by several orders of magnitude before settling at a very small value. This behaviour is consistent with the excellent agreement observed on the right-edge deflection. For clarity, only the most relevant loss components are shown in Fig. 8; the full loss history, including all auxiliary terms, is provided in Appendix A.

Overall, the loss evolution highlights a trade-off induced by the chosen loss weights: the training process prioritizes accurate satisfaction of the local PDE and boundary conditions, while global moment equilibrium is only matched approximately, even though the global compliance at the tip is recovered very accurately.

### 5.4 Qualitative Interpretation

Before discussing the spatial patterns of the fields, it is useful to summarise a few key beam-level quantities. Table 1 compares the tip deflection and maximum bending stress predicted by Euler–Bernoulli beam theory, the PINN, and the FEA reference solution. For stresses, absolute values are reported.

| Quantity | Theory | PINN | FEA |
|---|---|---|---|
| Tip deflection $v_{\mathrm{right}}$ [mm] | $-4.3643$ | $-4.3616$ | $-4.395$ |
| Max bending stress $\lvert\sigma_{xx,\max}\rvert$ [MPa] | 72.012 | 71.609 | 67.35 |

Table 1: Comparison of key beam-level quantities for Euler–Bernoulli theory, the PINN, and the FEA reference solution. For stresses, absolute values are reported.

From a mechanical standpoint, the PINN solution exhibits the correct qualitative behaviour of a cantilever beam in bending and compares favourably with the FE reference fields:

- the clamped boundary is effectively enforced, with both displacement components close to zero at $x = 0$ in both PINN and FE solutions;
- the transverse displacement field $u_y$ shows a smooth bending profile with curvature increasing

towards the clamped end. The tip deflection predicted by the PINN differs from the Euler–Bernoulli value by about 0.06%, while the FE solution deviates by about 0.7%, so that both models reproduce the global compliance very accurately, with the PINN slightly closer to the analytical prediction;

- the axial displacement $u_x$ exhibits the expected antisymmetric bending pattern with extension on one side of the beam and compression on the other. The axial-to-transverse displacement ratios are $\max|u_x|/\max|u_y| \approx 8.0 \times 10^{-2}$ for the PINN and $\approx 7.5 \times 10^{-2}$ for FEA, both somewhat lower than the slender-beam estimate $h/L = 1.0 \times 10^{-1}$, but in good mutual agreement and consistent with a bending-dominated response;

- the normal stress $\sigma_{xx}$ recovers the correct linear variation through the thickness and the expected tension/compression pattern at the clamped edge. In terms of magnitude, the peak bending stress differs from the Euler–Bernoulli prediction by about 0.6% for the PINN and 6.5% for the FE model, the latter slightly underestimating the maximum due to mesh resolution. In this sense, the PINN is actually closer to the theoretical value while remaining qualitatively consistent with the FE contours.

The main remaining discrepancies are a modest underestimation of the axial-to-transverse displacement ratio and a small *under*estimation of the peak bending stress in the PINN, contrasted with a more pronounced underestimation of this stress in the FE solution. This suggests that the current choice of loss weights favours local PDE satisfaction and global compliance at the tip over an exact match of the internal bending moment distribution. Nevertheless, the results demonstrate that a moderately sized PINN is already able to recover a mechanically consistent stress field, with errors of about 0.6% in maximum bending stress and 0.06% in tip deflection, making it a viable surrogate for this class of elasticity problems, complementary to classical FEA.

In summary, the trained PINN reproduces the cantilever response with a high level of accuracy at both the global and local levels. Beam-level quantities such as the tip deflection and maximum bending stress are within 0.06% and 0.6% of the Euler–Bernoulli predictions, respectively. At the field level, the displacement and stress contours are in very good qualitative agreement with the FEA solution, with noticeable discrepancies mainly confined to the vicinity of the clamped boundary and to the axial displacement component. A more detailed quantitative comparison with the reference FEA solution, based on relative error norms, is presented in Section 6.

# 6 Benchmarking Against FEA

## 6.1 Reference FEA Solution

To provide a numerical reference for the PINN predictions, the cantilever beam defined in Section 2 is solved using a linear static finite element model. The computation is performed in OptiStruct (Altair), with the command card `ANALYSIS = STATICS`, corresponding to a small-deformation, linear-elastic plane-stress analysis. The same geometry, material parameters and loading conditions as defined in Section 2 are used.

The beam is discretized using four-node quadrilateral elements (`CQUAD4`) under a plane-stress formulation. The mesh contains $N_x \times N_y = 100 \times 10$ elements, resulting in a typical element size of $\Delta x \approx 5$ mm along the span and $\Delta y \approx 5$ mm through the thickness. This discretization is fine enough to provide an accurate reference for global quantities (such as tip deflection and overall compliance), although local peak stresses remain somewhat mesh-dependent, as discussed in Section 5.2.

A fully clamped boundary condition is imposed on the left edge $x = 0$ by fixing all translational degrees of freedom of the nodes located on this boundary:

$$u_x = u_y = 0 \qquad \text{for all nodes with } x = 0.$$

In the OptiStruct input file, this corresponds to `SPC` entries that constrain the in-plane translations (and associated drilling rotation) to zero.

The external load consists of a vertical nodal force of magnitude $F = 120$ N applied downward at the free end of the beam. In the OptiStruct deck this is specified through a `FORCE` card with a load vector oriented in the negative $y$-direction.

From the converged FEA solution, nodal displacements $\{u_x^{\text{FEA}}, u_y^{\text{FEA}}\}$ are exported for all mesh nodes, along with Cauchy stress components $\{\sigma_{xx}^{\text{FEA}}, \sigma_{xy}^{\text{FEA}}\}$ evaluated at the element level. The axial stress $\sigma_{xx}^{\text{FEA}}$ is then post-processed into a nodal field by averaging the element values attached to each node (see Appendix B). These fields form the reference solution against which the PINN predictions are compared in Section 6.3.

## 6.2 Error Metrics

To quantitatively assess the agreement between the PINN and the reference FEA solution described in Section 6.1, the PINN is evaluated at the same spatial locations as the finite element nodes. Denoting by $(x_i, y_i)$, $i = 1, \dots, N$, the coordinates of the FEA mesh nodes, we obtain for each scalar field of interest $f$ the pair of discrete values

$$f_i^{\text{FEA}} = f^{\text{FEA}}(x_i, y_i), \qquad f_i^{\text{PINN}} = f^{\text{PINN}}(x_i, y_i), \qquad i = 1, \dots, N.$$

In this work, we focus on the in-plane displacement components $u_x, u_y$ and on the axial stress $\sigma_{xx}$, which are the most relevant quantities for bending of a cantilever beam. The FEA axial stress field $\sigma_{xx}^{\text{FEA}}$ is the nodal-averaged quantity obtained from the element stresses as described in Appendix B.

For any such scalar field $f \in \{u_x, u_y, \sigma_{xx}\}$, the *relative* discrete $L^2$ error between the PINN and the FEA reference is defined as

$$\varepsilon_{L^2}(f) = \frac{\left(\sum_{i=1}^{N} \left| f_i^{\mathrm{PINN}} - f_i^{\mathrm{FEA}} \right|^2\right)^{1/2}}{\left(\sum_{i=1}^{N} \left| f_i^{\mathrm{FEA}} \right|^2\right)^{1/2}}. \tag{6.1}$$

This quantity measures the global root-mean-square discrepancy between the two fields, normalized by the energy-like norm of the FEA solution. Values of $\varepsilon_{L^2}(f)$ close to zero indicate good overall agreement.

In addition, a relative discrete $L^\infty$ (maximum) error is considered:

$$\varepsilon_{L^\infty}(f) = \frac{\max_{1\le i\le N} \left| f_i^{\mathrm{PINN}} - f_i^{\mathrm{FEA}} \right|}{\max_{1\le i\le N} \left| f_i^{\mathrm{FEA}} \right|}. \tag{6.2}$$

This metric characterizes the worst pointwise deviation between the PINN prediction and the reference field, normalized by the maximum magnitude of the FEA solution.

Beyond these global norms, pointwise error fields are also considered to analyze the spatial distribution of the discrepancies. For each scalar quantity $f$, the pointwise error at node $i$ is defined as

$$e_f(x_i, y_i) = f_i^{\mathrm{PINN}} - f_i^{\mathrm{FEA}}, \qquad i = 1, \ldots, N, \tag{6.3}$$

and can be visualized as a colour map over the beam domain. Such plots make it possible to identify regions where the PINN closely follows the FEA solution, and regions where larger errors are concentrated (e.g. near the clamped boundary or in zones of high stress gradients). The corresponding field comparisons and error maps are presented in Section 6.3.

## 6.3 Comparison of Fields

This section compares the displacement and stress fields predicted by the PINN with the reference finite element (FEA) solution described in Section 6.1. All PINN quantities are evaluated at the exact coordinates of the FEA mesh nodes, allowing a pointwise comparison of $u_x$, $u_y$ and $\sigma_{xx}$.

Figures 3–5 show the FEA reference fields, while Figures 2–4 present the corresponding PINN predictions. The pointwise error maps

$$e_f(x_i, y_i) = f_i^{\mathrm{PINN}} - f_i^{\mathrm{FEA}}$$

for $u_x$, $u_y$ and $\sigma_{xx}$ are displayed in Figures 9 to 11.

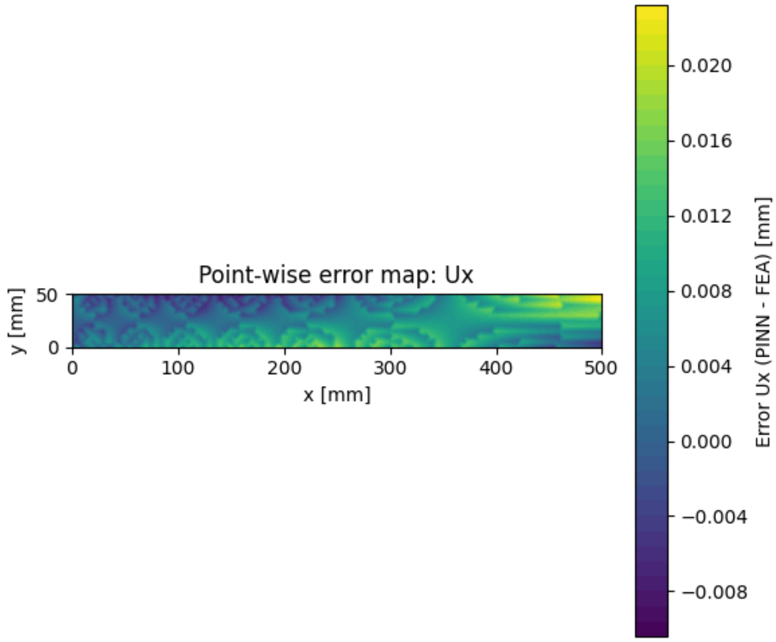


Figure 9: Pointwise error map for the axial displacement: $e_{u_x}(x, y) = u_x^{\mathrm{PINN}}(x, y) - u_x^{\mathrm{FEA}}(x, y)$, evaluated at the FEA nodes. The colour scale is in millimetres.

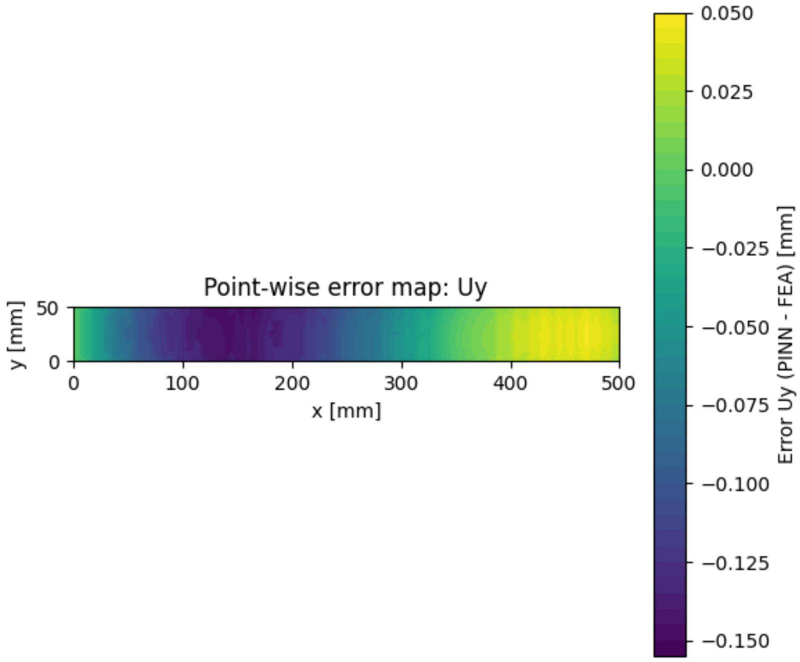


Figure 10: Pointwise error map for the vertical displacement: $e_{u_y}(x, y) = u_y^{\mathrm{PINN}}(x, y) - u_y^{\mathrm{FEA}}(x, y)$, evaluated at the FEA nodes. The colour scale is in millimetres.

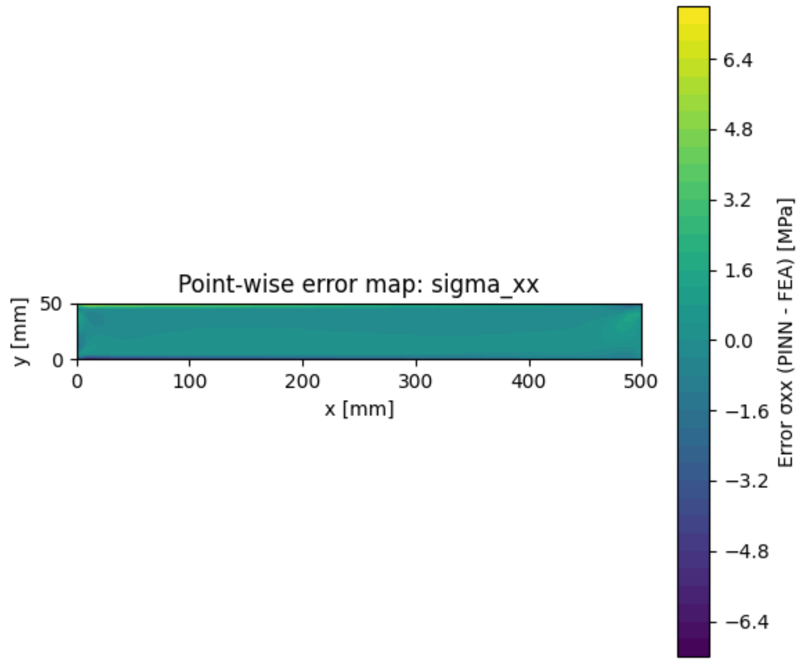


Figure 11: Pointwise error map for the axial stress: $e_{\sigma_{xx}}(x, y) = \sigma_{xx}^{\mathrm{PINN}}(x, y) - \sigma_{xx}^{\mathrm{FEA}}(x, y)$, evaluated at the FEA nodes. Units are MPa.

Qualitatively, the displacement fields exhibit the expected bending pattern: $u_x$ varies almost linearly across the thickness, while $u_y$ forms smooth arcs from the clamped boundary to the free end. The stress field $\sigma_{xx}$ presents the characteristic linear-through-thickness bending distribution, with maximum tension and compression localised at the clamped section.

The pointwise error fields confirm the good overall agreement between PINN and FEA. The axial displacement error remains below 0.01 mm over most of the domain, with no significant drift along the beam, and reaches local peaks of order $2 \times 10^{-2}$ mm near the clamped edge. The vertical deflection error is similarly small, typically within 0.12 mm and with a maximum amplitude of about 0.15 mm. The spatial pattern of the $u_y$ error indicates that the PINN slightly underestimates the deflection in the central region and slightly overestimates it near the free end, consistent with a minor bias in the curvature.

The stress error is localized primarily near the clamped boundary, where stress gradients are largest. Close to this region, the error reaches values of order ±6–7 MPa, i.e. roughly 9–10% of the peak bending stress. Away from the fixed support, the $\sigma_{xx}$ error remains typically below ±3–4 MPa, corresponding to relative variations of only a few percent. This behaviour is expected: enforcing local PDE residuals in a strong-form PINN tends to smooth high-frequency stress variations, whereas the FEA solution contains sharper gradients inherited from the finite element shape functions.

### 6.4 Discussion of Discrepancies

The quantitative error measures introduced in Section 6.2 confirm the good overall accuracy of the PINN solution. The relative $L^2$ errors between PINN and FEA read

$$\varepsilon_{L^2}(u_x) = 5.08 \times 10^{-2}, \qquad \varepsilon_{L^2}(u_y) = 4.08 \times 10^{-2}, \qquad \varepsilon_{L^2}(\sigma_{xx}) = 6.84 \times 10^{-2},$$

while the corresponding relative $L^\infty$ errors are

$$\varepsilon_{L^\infty}(u_x) = 6.94 \times 10^{-2}, \qquad \varepsilon_{L^\infty}(u_y) = 3.43 \times 10^{-2}, \qquad \varepsilon_{L^\infty}(\sigma_{xx}) = 1.10 \times 10^{-1}.$$

In other words, the global discrepancies remain in the range of 4–7% in the $L^2$ norm, with maximum pointwise deviations of about 7% for the displacements and roughly 11% for the axial stress.

For clarity, these error values are summarised in Table 2.

| Quantity | $\varepsilon_{L^2}$ | $\varepsilon_{L^\infty}$ |
|---|---|---|
| $u_x$ | $5.08 \times 10^{-2}$ | $6.94 \times 10^{-2}$ |
| $u_y$ | $4.08 \times 10^{-2}$ | $3.43 \times 10^{-2}$ |
| $\sigma_{xx}$ | $6.84 \times 10^{-2}$ | $1.10 \times 10^{-1}$ |

Table 2: Relative $L^2$ and $L^\infty$ errors between the PINN prediction and the FEA reference solution for the main fields of interest.

From a PINN standpoint, these error levels are reasonable: achieving relative $L^2$ errors of a few percent on displacements and stresses, together with maximum relative errors of about 7% on the displacements and 11% on $\sigma_{xx}$, is typical for strong-form PINNs in 2D elasticity when no displacement or stress data are provided during training and only PDE residuals and global constraints are enforced. In this sense, the present model sits in the lower range of reported displacement errors, while the stress discrepancies remain dominated by the high-gradient region near the clamped boundary and the smoothing nature of the neural approximation.

Three main observations can be made from the pointwise error fields:

**(i) Displacement accuracy:** Both components of the displacement field are well reproduced. The error in $u_x$ remains below 0.01 mm almost everywhere, with local peaks of order $10^{-2}$ mm near the clamped boundary. The $u_y$ error shows a smooth spatial pattern along the beam axis: slightly more negative than the FEA prediction in the mid-span region, and slightly less negative near the free end. This behaviour is consistent with a small curvature bias induced by the interplay of global constraints (tip deflection requirement) and local PDE residuals. Importantly, the magnitude of the error remains extremely small relative to the physical displacement ($v_{\text{tip}} \approx -4.36$ mm).

**(ii) Stress discrepancies near the clamped boundary:** The stress field exhibits larger errors close to the fixed support, where $\sigma_{xx}$ changes rapidly through the thickness. This is expected for strong-form PINNs, which favour smooth solutions and may struggle to reproduce sharp gradients unless densely sampled near those regions. Away from the clamped edge, the stress error becomes almost uniform and remains within a few MPa, i.e. less than about 5% of the stress magnitude.

**(iii) Local oscillations in $u_x$:** The axial displacement error displays mild oscillations near the clamped boundary. These oscillations are attributable to two combined effects: (i) the low sensitivity of $u_x$ to the applied loading compared to $u_y$, which makes $u_x$ more difficult to resolve from PDE residuals alone, and (ii) the absence of explicit data or regularization on transverse derivatives, which can lead to small non-physical modes being introduced during training. Such artifacts are commonly reported in strong-form PINNs for elasticity, and their amplitude remains very small (of order $10^{-2}$ mm).

Overall, the discrepancies remain moderate and localized. The PINN correctly captures the global bending behaviour, the stress distribution, and the tip deflection. The observed errors are consistent with the known limitations of strong-form PINNs in solid mechanics: sensitivity to boundary layers, high-gradient regions, and the balance between local PDE residuals and global constraints.

# 7 Data-Assisted Extension of the PINN

In the previous sections, the PINN was trained in a purely physics-informed manner, using only PDE residuals, boundary conditions and beam-level constraints in the loss function. This section investigates whether a small amount of data from the finite element (FEA) reference solution can be used to refine the prediction, and in particular to further reduce the discrepancies on the axial stress field $\sigma_{xx}$. To this end, a second training stage is carried out in which the physics-only PINN is fine-tuned using a hybrid physics-and-data loss.

## 7.1 Hybrid Physics-and-Data Loss Formulation

Let $\mathbf{u}_\theta = (u_\theta, v_\theta)$ denote the displacement field predicted by the PINN with parameters $\theta$, and let $\sigma_{xx,\theta}$ be the corresponding axial stress field recovered via the linear elastic constitutive law as in Section 5.2. In the physics-only stage, the model is trained by minimizing the total loss

$$\begin{aligned} L^{(1)}_{\text{total}} = {} & W_{\text{PDE}} L_{\text{PDE}} + W_{\text{BC}} L_{\text{BC}} + W_{\text{res}} L_{\text{res}} + W_{\text{free}} L_{\text{free}} \\ & + W_{\text{neutral}} L_{\text{neutral}} + W_{\text{moment}} L_{\text{moment}} + W_{\text{beam}} L_{\text{beam}} \\ & + W_{\text{unif}} L_{\text{unif}} + W_{\text{tip}} L_{\text{tip}}. \end{aligned} \tag{7.1}$$

where the individual components are defined in Section 4 and the weights are given in the numerical setup of Section 5.3.

Starting from the best physics-only parameters, a second *data-assisted* stage is performed in which the loss function is augmented with a term that penalises discrepancies between the PINN predictions and the FEA fields at the mesh nodes. Denoting by $(x_j, y_j)$, $j = 1, \ldots, N_{\text{data}}$, the coordinates of the FEA nodes for which both displacements and nodal-averaged axial stress are available (see Section 6.1 and Appendix B), the data misfit is defined as

$$L_{\text{data}} = \frac{1}{N_{\text{data}}} \sum_{j=1}^{N_{\text{data}}} \Big( \big\| u_\theta(x_j, y_j) - u_x^{\text{FEA}}(x_j, y_j) \big\|^2 + \big\| v_\theta(x_j, y_j) - u_y^{\text{FEA}}(x_j, y_j) \big\|^2 + \big\| \sigma_{xx,\theta}(x_j, y_j) - \sigma_{xx}^{\text{FEA}}(x_j, y_j) \big\|^2 \Big). \tag{7.2}$$

In the second stage, the total loss is replaced by

$$L_{\text{total}}^{(2)} = W_{\text{PDE}} L_{\text{PDE}} + W_{\text{BC}} L_{\text{BC}} + W_{\text{tip}} L_{\text{tip}} + W_{\text{data}} L_{\text{data}}, \tag{7.3}$$

where the weights $W_{\text{PDE}}$, $W_{\text{BC}}$ and $W_{\text{tip}}$ are kept identical to their physics-only values, while a single scalar parameter $W_{\text{data}}$ controls the relative importance of the FEA fields in the hybrid loss. The other auxiliary physics terms (moment balance, neutral axis, beam prior, etc.) are not used in this fine-tuning stage, so that the effect of the data term can be isolated more clearly.

The data-assisted PINN is thus obtained by minimizing $L_{\text{total}}^{(2)}$ with respect to $\theta$, starting from the parameters that minimize $L_{\text{total}}^{(1)}$.

## 7.2 Selection of FEA Training Samples

The FEA model described in Section 6.1 provides nodal displacements and nodal-averaged axial stresses on the structured $100 \times 10$ element mesh of the cantilever beam. For the data-assisted stage, the FEA training set is constructed as follows:

- the node IDs present in the OptiStruct output are matched to the node IDs used in the PINN post-processing, ensuring a one-to-one correspondence between $(x_j, y_j)$ and the PINN evaluation points;
- only nodes for which all three quantities $(u_x^{\text{FEA}}, u_y^{\text{FEA}}, \sigma_{xx}^{\text{FEA}})$ are available are retained, discarding entries with missing values;
- all remaining nodes are used in the loss (7.2), so that the FEA data cover the full span and thickness of the beam, including the high-gradient region near the clamped boundary.

No explicit oversampling of the clamped edge is introduced in this first experiment; instead, the effect of the data term is assessed using the existing FEA sampling inherited from the finite element mesh. This choice keeps the hybrid formulation simple and mirrors the evaluation grid used in Section 6.3 for the error metrics.

## 7.3 Impact on Beam-Level Quantities

As a first consistency check, the data-assisted PINN is evaluated on the beam-level quantities introduced in Section 5.4, namely the right-edge deflection $v_{\text{right}}$ and the maximum axial stress at the clamped edge $\sigma_{xx,\max}$. The corresponding values remain extremely close to those reported in

Table 1: the right-edge deflection differs only marginally from the Euler–Bernoulli prediction, and the peak bending stress at the encastrement stays within a few percent of the theoretical value.

This behaviour is expected. Since the tip-deflection constraint $L_{\mathrm{tip}}$ and the PDE residual $L_{\mathrm{PDE}}$ are retained in the hybrid loss (7.3), the global compliance and bending moment distribution are still controlled primarily by the physics-based terms. The additional data term mainly acts as a local correction on the displacement and stress fields, without significantly altering the global response of the beam. In particular, the data-assisted PINN remains very close to both the Euler–Bernoulli solution and the physics-only PINN at the beam level.

## 7.4 Improvement of Local Field Accuracy

The impact of the data-assisted stage on the full fields is quantified using the relative $L^2$ and $L^\infty$ errors defined in Eqs. (6.1)–(6.2). Denoting by $\varepsilon_{L^2}^{\mathrm{phys}}(f)$ and $\varepsilon_{L^\infty}^{\mathrm{phys}}(f)$ the errors of the physics-only model, and by $\varepsilon_{L^2}^{\mathrm{data}}(f)$ and $\varepsilon_{L^\infty}^{\mathrm{data}}(f)$ those of the data-assisted PINN, the numerical values obtained for $f \in \{u_x, u_y, \sigma_{xx}\}$ are summarised in Table 3.

| Quantity | Metric | Physics-only PINN | Data-assisted PINN |
|---|---|---|---|
| $u_x$ | $\varepsilon_{L^2}$ | $5.08 \times 10^{-2}$ | $6.49 \times 10^{-2}$ |
| $u_y$ | $\varepsilon_{L^2}$ | $4.08 \times 10^{-2}$ | $4.25 \times 10^{-2}$ |
| $\sigma_{xx}$ | $\varepsilon_{L^2}$ | $6.84 \times 10^{-2}$ | $5.27 \times 10^{-2}$ |
| $u_x$ | $\varepsilon_{L^\infty}$ | $6.94 \times 10^{-2}$ | $8.13 \times 10^{-2}$ |
| $u_y$ | $\varepsilon_{L^\infty}$ | $3.43 \times 10^{-2}$ | $3.55 \times 10^{-2}$ |
| $\sigma_{xx}$ | $\varepsilon_{L^\infty}$ | $1.10 \times 10^{-1}$ | $6.72 \times 10^{-2}$ |

Table 3: Relative $L^2$ and $L^\infty$ errors between the PINN predictions and the FEA reference solution for the physics-only and data-assisted models.

The displacement errors are only mildly affected by the data-assisted stage. For both components, the relative $L^2$ and $L^\infty$ errors of the hybrid model remain in the same range as those of the physics-only PINN, with variations of at most a few percentage points. This confirms that the physics-only training already produces a displacement field that closely matches the FEA reference.

In contrast, the agreement on the axial stress field $\sigma_{xx}$ is noticeably improved. The relative $L^2$ error decreases from $\varepsilon_{L^2}^{\mathrm{phys}}(\sigma_{xx}) \approx 6.84\times10^{-2}$ to $\varepsilon_{L^2}^{\mathrm{data}}(\sigma_{xx}) \approx 5.27\times10^{-2}$, and the maximum relative error drops from $\varepsilon_{L^\infty}^{\mathrm{phys}}(\sigma_{xx}) \approx 1.10 \times 10^{-1}$ to $\varepsilon_{L^\infty}^{\mathrm{data}}(\sigma_{xx}) \approx 6.72 \times 10^{-2}$. The data-assisted PINN thus reduces the worst pointwise discrepancy on $\sigma_{xx}$ by almost a factor of two, while preserving the overall level of accuracy on the displacements.

These results indicate that incorporating a moderate amount of FEA data on both displacements and stresses primarily helps to correct local stress discrepancies, especially in high-gradient regions near the clamped boundary and along the outer fibres, without degrading the global mechanical response.

### 7.5 Discussion and Perspectives

The data-assisted experiment highlights the genuinely multi-objective nature of training PINNs for elasticity problems. A preliminary test (not reported in detail here) in which the data term was applied only to the displacement components $u_x$ and $u_y$ led to a marked reduction in the displacement errors, but at the cost of significantly larger discrepancies in $\sigma_{xx}$. In other words, supervising displacements alone is not sufficient to guarantee an accurate reconstruction of the stress field, which depends on spatial derivatives of $\mathbf{u}_\theta$ and on the balance of internal and external loads.

By including the axial stress in the data loss (7.2) and retaining the key physics-based constraints (equilibrium PDE and tip deflection), the hybrid formulation recovers a more favourable compromise: the displacement errors remain essentially unchanged compared with the physics-only PINN, while the global and maximum errors on $\sigma_{xx}$ are significantly reduced. This suggests that, for solid mechanics applications, data-assisted PINNs should, whenever possible, incorporate information on both displacements and stresses (or strains), rather than focusing solely on the kinematic fields.

From a broader perspective, the present results illustrate how a modest amount of high-fidelity FEA data can be used to refine a physics-informed surrogate without sacrificing its mechanical consistency or its agreement with beam theory. In future work, similar hybrid strategies could be applied to more complex geometries, three-dimensional elasticity problems, or multi-physics settings, and could be combined with alternative surrogate architectures such as graph neural networks for mesh-based data, as discussed in Section 9.3.

## 8 Discussion

### 8.1 Why the Physics-Only PINN Works

The results of Sections 5–6 show that the physics-only PINN already reproduces the cantilever response with a high level of accuracy. Beam-level quantities such as the tip deflection and the maximum bending stress are within 0.06% and 0.6% of the Euler–Bernoulli predictions, respectively, and the displacement and stress fields match the FEA reference with relative $L^2$ errors in the range of 4–7%.

Several factors contribute to this strong performance. First, the mechanical problem considered here is essentially linear and close to the classical beam-theory regime: the deformation is dominated by bending, the geometry is simple, and the boundary conditions are well posed. In such a setting, the equilibrium PDE, traction-free conditions and clamped boundary at the left edge already contain enough information to largely determine the shape of the displacement field.

Second, the loss function used in the physics-only stage combines local and global constraints. The PDE residual and boundary-condition terms enforce pointwise equilibrium and compatibility across the domain, while the beam-level terms (moment balance, beam prior, tip deflection, neutral axis, etc.) encode integral properties of the solution and anchor the overall amplitude of the response. This multi-scale loss design effectively regularises the training and steers the PINN towards a mechanically consistent solution, even in the absence of any observational data.

Third, the neural network architecture is sufficiently expressive to represent the smooth displacement and stress fields of the cantilever. Once the loss weights have been tuned to a reasonable balance, the optimiser finds a set of parameters that jointly satisfy the different constraints to a good degree. The resulting model therefore behaves as a purely physics-based surrogate, producing accurate predictions throughout the domain without having seen a single FEA sample during training.

### 8.2 Effect of Data-Assisted Fine-Tuning

The data-assisted stage of Section 7.2 starts from the physics-only solution and fine-tunes the parameters using a hybrid loss that combines the PDE residual, clamped boundary conditions, tip-deflection constraint and a data term involving nodal displacements and axial stresses from FEA. This procedure yields a PINN that is still constrained by the governing equations, but additionally calibrated on a modest amount of high-fidelity data.

At the beam level, the impact of this fine-tuning is intentionally limited. The right-edge deflection and peak bending stress remain extremely close to both the Euler–Bernoulli solution and the physics-only PINN, confirming that the global response is still governed primarily by the physics-based terms in the loss. This behaviour is desirable: the data term is used as a correction on local field details rather than as a replacement for the underlying mechanical model.

The effect is more visible at the field level, especially for the axial stress $\sigma_{xx}$. While the relative $L^2$ and $L^\infty$ errors on the displacements change only marginally between the two stages, the discrepancies on $\sigma_{xx}$ are significantly reduced. The global relative error decreases from about 6.8% to 5.3%, and the maximum pointwise error drops from approximately 11% to 6.7%. In other words, the data-assisted PINN nearly halves the worst local error on the stress field, while preserving the excellent agreement on displacements and beam-level quantities.

This behaviour reflects the multi-objective nature of the hybrid training. With a moderate value of the data weight, the PDE, boundary and tip-deflection terms continue to control the overall deformation pattern and compliance, whereas the FEA samples act mainly to correct local stress discrepancies, especially in high-gradient regions near the clamped edge and along the outer fibres. The resulting compromise is favourable for applications where accurate local stresses are required but only a limited amount of simulation data is available.

### 8.3 Strong-Form PINN Limitations in High-Gradient Regions

Despite the generally good agreement with FEA, the error maps of Section 6.3 and the metrics of Section 6.4 reveal systematic discrepancies near the fixed boundary and in regions where stress gradients are large. These patterns are consistent with known limitations of strong-form PINNs for elasticity problems.

In a strong-form formulation, the equilibrium equations are enforced pointwise at a finite set of collocation points using automatic differentiation. This approach tends to favour smooth solutions and may struggle to capture sharp variations unless the sampling is strongly concentrated in high-gradient areas. In the present work, the collocation points are distributed relatively uniformly across the domain, without explicit oversampling of the clamped edge. As a result, the PINN under-resolves the steep thickness-wise variation of $\sigma_{xx}$ in the immediate vicinity of the encastrement, leading to local errors of the order of 10% in the physics-only stage.

Moreover, the stress field is obtained from spatial derivatives of the displacements through the constitutive law. Any small inaccuracy or high-frequency oscillation in $\mathbf{u}_\theta$ is therefore amplified when computing $\sigma_{xx}$, especially near boundaries where the gradients are largest. This explains why the relative errors on $\sigma_{xx}$ are systematically higher than those on $u_x$ and $u_y$, even though the displacement errors themselves remain very small.

The multi-term loss also induces potential conflicts between objectives: local PDE residuals, boundary conditions, moment balance, tip deflection and data misfit all pull the network in slightly different directions in regions where the FEA solution does not satisfy the idealised beam assumptions exactly. In such cases, the optimiser converges to a compromise that matches the main constraints well, but cannot simultaneously minimise every loss term in high-gradient zones.

The data-assisted stage partly alleviates these issues by injecting stress information directly into the loss, thereby guiding the network towards a more accurate stress profile near the clamped edge. However, the underlying limitations of the strong-form approach remain: the method is sensitive to the distribution of collocation points, the choice of loss weights and the smoothness of the network. Alternative formulations based on weak forms or energy principles, or adaptive sampling strategies that enrich the collocation set where residuals are large, could further improve the accuracy in boundary layers and gradient-dominated regions.

## 8.4 Role of PINNs Relative to Classical FEA

The comparison with the OptiStruct reference solution provides a concrete perspective on how PINNs position themselves with respect to classical finite element analysis. In the present single-load, single-geometry setting, FEA is clearly the most efficient tool for obtaining a highly accurate solution: once the mesh is built, the linear elastic problem can be solved deterministically with well-controlled discretisation errors, and local quantities such as stresses can be extracted with mature post-processing techniques.

The PINN, by contrast, incurs a substantial upfront training cost to reach a solution that is comparable in accuracy to FEA. From a purely computational standpoint, this is not advantageous for a one-off analysis. However, the trained network offers different capabilities: it provides a mesh-free, differentiable surrogate of the displacement field over the entire domain, from which stresses and other derived quantities can be obtained at arbitrary points. Once trained, evaluating the PINN is inexpensive, and the model can be queried repeatedly without rerunning a finite element solve.

In its current form, the PINN is tailored to a single boundary-value problem. To fully exploit its surrogate nature, one would need to extend the input space to include parameters such as the applied load, material properties or geometric dimensions, so that a single trained network could approximate an entire family of FEA solutions. In that regime, PINNs and related physics-informed surrogates become attractive complements to FEA for parametric studies, optimisation loops or real-time applications where thousands of forward evaluations are required.

The present study therefore should not be viewed as a competition between PINNs and finite elements, but rather as a demonstration that a moderately sized, carefully trained PINN can reproduce a standard elasticity benchmark with errors of only a few percent, even in a physics-only setting, and that a small amount of high-fidelity data further improves the stress prediction. Classical FEA remains the primary tool for routine structural analysis, while PINNs provide a promising avenue for reduced-order, differentiable and data-enhanced surrogates. These roles are complementary, and their combination is likely to be particularly powerful in more complex settings, as further discussed in Section 9.3.

# 9 Conclusion

## 9.1 Summary of Main Findings

This work has investigated the use of Physics-Informed Neural Networks to approximate the plane-stress response of a cantilever beam and has benchmarked the resulting surrogate against a finite element reference solution. The physics-only PINN, trained solely on equilibrium equations, boundary conditions and beam-level constraints, already reproduces the global response with very high accuracy: the beam-level quantities and displacement fields are within a few percent of both Euler–Bernoulli beam theory and the OptiStruct solution, as detailed in Sections 5–6.

Building on this baseline, a second, data-assisted training stage fine-tunes the network using a hybrid loss that combines physics terms with a misfit to nodal displacements and stresses from FEA. This hybrid formulation leaves the global response essentially unchanged, while noticeably reducing the discrepancies on the axial stress field, in particular near the clamped edge. Overall, the study shows that moderately sized PINNs with carefully designed physics-based losses can act as accurate surrogates for standard elasticity benchmarks, and that a small amount of simulation data can be used to refine local stresses without sacrificing mechanical consistency.

## 9.2 Practical Limitations and Scope of the Approach

The present results remain constrained by several practical limitations. The benchmark problem is linear, involves a simple rectangular geometry and a single load case, and lies well within the validity range of classical beam theory. Moreover, the strong-form formulation used here is sensitive to the distribution of collocation points and the relative weighting of the different loss terms, and its training cost is significantly higher than that of a single finite element solve. As discussed in

Section 8.3, these factors restrict the current PINN to smooth, well-understood problems where the cost of an initial training phase can be amortised over many subsequent evaluations.

Within this scope, the model should therefore be viewed not as a replacement for classical finite element analysis, but as a proof-of-concept surrogate for parametric studies, optimisation loops or digital-twin scenarios in which repeated, inexpensive evaluations of a differentiable approximation are advantageous.

## 9.3 Future Work: From PINNs to GNN-Based Surrogates

The methodology explored in this report opens several directions for future work. A first step is to move from single-instance surrogates to parameterised models that take as input not only the spatial coordinates, but also loading and material parameters, so that a single network can approximate an entire family of finite element solutions.

A second direction is to combine physics-informed training with mesh-based architectures such as graph neural networks (GNNs). Unlike the fully connected PINN used here, GNNs operate directly on the finite element mesh, propagate information along element connectivity and can naturally handle complex geometries and non-uniform refinements. Embedding equilibrium constraints and constitutive relations into such graph-based surrogates would make it possible to leverage both the structure of the mesh and the expressiveness of modern geometric deep learning.

Finally, the hybrid strategy tested in this work could be generalised to three-dimensional and multi-physics problems, where high-fidelity FEA data are expensive but partial information on displacements, strains or stresses is available. In that context, combining physics-based losses, limited simulation data and advanced architectures such as GNNs offers a promising route towards scalable, mechanically consistent surrogates for industrial virtual testing and design.

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# A Full Loss History

This appendix reports the complete loss history of the PINN training, including all auxiliary terms that are not shown in Fig. 8 for readability. In addition to the total loss and the main components ($L_{\text{PDE}}$, free-edge boundary term, moment balance, tip-deflection constraint), the figure below displays the evolution of the plane-stress prior on $\sigma_{yy}$, the neutral-axis constraint, the midline $\sigma_{xx}$ term, the beam-theory prior, and the right-edge uniformity loss.

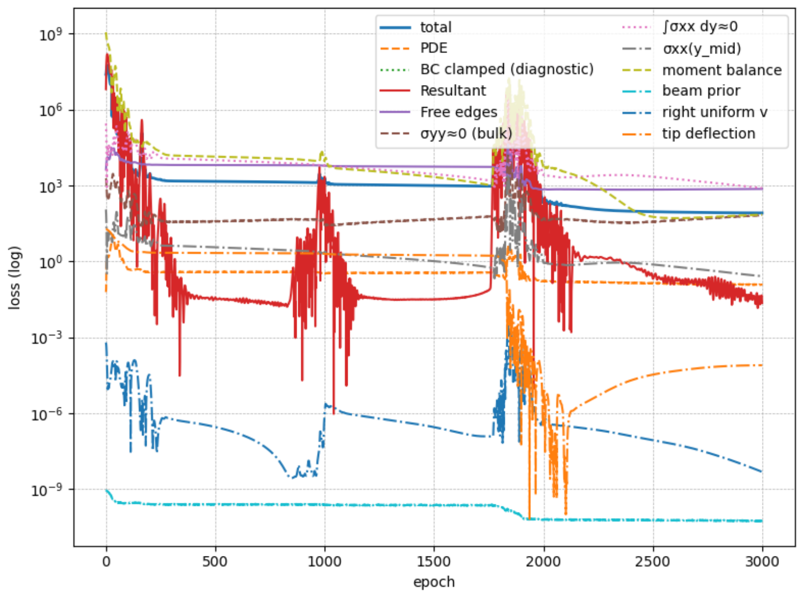


Figure 12: Complete convergence history of all loss components used during training of the PINN, plotted on a logarithmic scale.

# B Computation of Nodal Axial Stress from FEA Output

The finite element solver (OptiStruct) provides Cauchy stresses at the element level, with one value per quadrilateral element for the field reported as `ElementStresses2D&3D(XX_Max)`. To compare these stresses with the PINN prediction, a nodal field $\sigma_{xx}(x_i, y_i)$ is required. Since OptiStruct does not export nodal stresses by default for this result, a nodal averaging procedure was performed in Python.

Each CSV line contains:

- the element identifier,
- the scalar stress value $\sigma_{xx}^{(e)}$ for that element,
- the four node IDs associated with the CQUAD4 element.

The nodal stress field is computed by accumulating all element stresses shared by a node and taking the arithmetic average. The Python script used is:

```python
from collections import defaultdict
import pandas as pd

df = pd.read_csv("fea_element_stresses.csv") # contains element_id, sigma_xx, node1..node4

node_stress = defaultdict(list)

# accumulate stress contributions from all attached elements
for _, row in df.iterrows():
    sigma_elem = row["ElementStresses2D&3DXX_Max_GlobalSystem_f"]
    nodes = [row["node1"], row["node2"], row["node3"], row["node4"]]
    for n in nodes:
        node_stress[int(n)].append(float(sigma_elem))

# average per node
node_sigma = {
    node_id: sum(values)/len(values)
    for node_id, values in node_stress.items()
}

df_nodes = pd.DataFrame({
    "node_id": list(node_sigma.keys()),
    "sigma_xx": list(node_sigma.values())
})

df_nodes.to_csv("fea_sigma_xx_nodal_avg.csv", index=False)
```

This produces a nodal field suitable for pointwise comparison with the PINN prediction and for constructing the error map shown in Fig. 11.